

Set up with Anaconda

For students who already use Anaconda or Miniconda on Windows, macOS or Linux.

What you are setting up: an *environment* is a separate Python installation with its own packages. Conda creates and activates the course environment, named `esbmtk-practicals`. JupyterLab is the interface you open in your browser; a *kernel* is the Python process that runs notebook cells. Your R installation stays unchanged. You do not need `uv` or another Python installer.

1 Download the course and open a terminal

Download the complete course folder from your instructor and extract the ZIP. On Windows, right-click it and choose **Extract All**. On macOS, double-click it. Work in the extracted folder, not inside the ZIP.

The folder must contain **environment-anaconda.yml**, **scripts**, **notebooks** and **data**, together with the other supplied course files. The YAML file is a package recipe supplied by your instructor; you do not need to write it.

Windows: click Start or the Windows search bar, type **Anaconda Prompt**, and open it. This terminal is already prepared to use Conda. A terminal is simply a window where you type commands. You do not need to open Navigator first.

macOS: press **Command+Space**, type **Terminal**, and press Return. **Linux:** open Terminal (often Ctrl+Alt+T). Use a terminal already configured for your Anaconda or Miniconda installation.

Type this command and press Enter:

```
conda --version
```

A version number confirms Conda is available. If it is not recognised, use Anaconda Prompt on Windows. On macOS/Linux, ask your instructor to help connect your existing Conda installation to the terminal.

2 Know where to type commands

Run terminal commands **one line at a time**, pressing Enter and waiting for each to finish. They are not R code and do not go in RStudio or a notebook cell. Do not copy the prompt before a command, such as `(base) C:\Users\Alex>`.

The first installation needs internet access. **Pilot status:** package resolution was checked on Windows/Python 3.14; a fresh Anaconda installation and cross-platform pilots remain outstanding. The separate `uv` route has its own verification record. Reference: [Conda environments](#).

Install and open the course

Keep using the Anaconda Prompt or Terminal window from page 1.

3 Tell the terminal which folder to use

The terminal's **current folder** can differ from the folder shown in your file browser. `cd` means change directory (folder). If Alex extracted the course into Downloads, the command would be:

Windows Anaconda Prompt: `/d` also changes the drive if needed.

```
cd /d "C:\Users\Alex\Downloads\ESBMTK-practicals"
```

macOS / Linux:

```
cd "$HOME/Downloads/ESBMTK-practicals"
```

Use your actual folder name and location, including a suffix such as `-main`. On Windows, open the folder in File Explorer, click its address bar and copy that path between the quotes after `cd /d`.

Check the folder: on Windows, run `cd` by itself to print the location, then `dir` to list files. On macOS/Linux, use `pwd`, then `ls`. Find **environment-anaconda.yml**, **scripts**, **notebooks** and **data** before continuing. “From the course folder” means this terminal location.

4 Create, activate and check the environment

```
conda env create -f environment-anaconda.yml
conda activate esbmtk-practicals
python -m pip check
python scripts/check_environment.py
```

Creation downloads Python and the course packages; wait for it to finish. Activation selects them for this terminal; the prompt should show **(esbmtk-practicals)**. If this course environment already exists, skip creation and run the other three commands.

Expect **No broken requirements found**, then **Environment check passed**. **Next: start JupyterLab**. The second check tests numerical calculations, chemistry, workbook access and plotting. Stop and seek help if either check fails.

5 Register the course kernel and start JupyterLab

Stay in this activated terminal and course folder. Register the kernel once:

```
python -m ipykernel install --sys-prefix --name esbmtk-practicals
python -m jupyterlab
```

JupyterLab opens in your browser. Keep the terminal open; its prompt will not return while the server runs. If no browser opens, copy the local `http://localhost:.../lab` or `http://127.0.0.1:.../lab` link printed there into your browser. Keep the full link private.

Check success and start working

The steps below take place in JupyterLab, then in the terminal when you finish.

6 Confirm that JupyterLab uses the course Python

Click **File > New > Notebook** and choose **esbmtk-practicals** (or **ESBMTK practicals** if registered with the earlier guide). Paste this code into the first cell, then press **Shift+Enter**:

```
import sys
import numpy as np
import esbmtk
import PyCO2SYS
print(sys.executable)
np.testing.assert_allclose(np.linalg.solve(np.eye(2), np.ones(2)), [1, 1])
print("Setup check passed")
```

The Python path should identify the esbmtk-practicals environment and match the interpreter printed by the terminal check. Expect **Setup check passed**. If both terminal checks and this notebook check passed, **setup is complete**. You do not repeat creation or kernel registration each time.

Open or create notebooks while JupyterLab is running

Use the left file browser to open **notebooks/student/**, then double-click a course notebook. If it asks for the instructor's ESBMTK314 kernel, choose your course kernel instead. Create another notebook via **File > New > Notebook**. No terminal command is needed for each notebook. Save with Ctrl+S (Cmd+S on macOS). Marked exercise placeholders are intentional.

When you finish: save and shut down

Save your notebooks. Return to the terminal running JupyterLab, press **Ctrl+C**, and, if asked to shut down, type **y** and press Enter. Wait for the normal prompt to return, then close the browser tab and terminal. This stops notebook kernels: variables in memory are lost, but saved notebooks remain on disk.

Closing the browser tab alone does not normally stop JupyterLab. Reopen the local link still shown in its terminal if you only closed the tab; you do not need another server.

Next time: open JupyterLab again

Open Anaconda Prompt or your Conda-enabled Terminal, repeat `cd` to the course folder (page 2), then run:

```
conda activate esbmtk-practicals
python -m jupyterlab
```

Do this only when the server has stopped, including after restarting your laptop. Open your saved notebooks and rerun the cells needed to recreate variables.

If a step fails

File not found: check the terminal folder (page 2). **Wrong Python or missing package:** activate the course environment, launch Jupyter there and select its kernel. **Windows Python crash:** use the activated prompt. For other errors, send your instructor the full error text, command and operating system.